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# Adversarial Evaluation and Hardening of GPU Fingerprinting for Location Verification

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**Abstract**

Delay-based location verification protocols for high-performance AI accelerators require a mechanism to confirm that challenge responses originate from the specific physical GPU under audit rather than a decoy chip or replay cache. Prior work by Tee and Happel (2026) introduced a software-based GPU fingerprinting proof-of-concept using atomic memory contention across streaming multiprocessors (SMs), reporting up to 100% closed-set re-identification accuracy. However, closed-set re-identification does not establish adversarial authentication security when an open-set verifier must simultaneously accept genuine responses across collections and reject modeled spoofs. In this paper, we evaluate both the legacy GPU fingerprinting function across four collections on eight NVIDIA H200 GPUs (10,240 fingerprints) and a hardened probe architecture in a matched eight-H200 head-to-head trial (3,456 total captures). First, we show that the legacy challenge schedule exhibits a 6-bit linear congruential generator (LCG) modulo bottleneck (`lcg_state % 64`, permitting up to 91.7% related-seed replay acceptance) and vulnerability to coordinate-wise median offset modeling, allowing a seven-decoy H200 ensemble to reach 87.5% (224/256) acceptance at the pooled p95 cutoff. Second, we evaluate a hardened probe combining 20-round ChaCha20 challenge expansion, sleep-free 16-address contention, per-round SM rank permutations, and two-of-three micro-burst voting. On the matched eight-H200 cohort at pooled p95, the hardened probe reduces seven-decoy ensemble acceptance from 87.5% (224/256) to 9.8% (25/256) and achieves 0/256 (0.0%) for the predetermined single-decoy assignment (down from 138/256, 53.9%) at comparable genuine acceptance (219/256 [85.5%] vs. 221/256 [86.3%]). In a separate matched two-H200 NVL run, the hardened implementation reduced median cold process runtime by 36% (772 ms to 492 ms; 38.3 ms warm burst in a single-H200 NVL pilot). Third, in an exploratory offline follow-up inspecting the single target (GPU C) that accounted for 19 of the 25 remaining seven-decoy accepts cohort-wide (19/32 targeting C), we observe that accepted ensemble bursts scored below every genuine GPU C burst on the burst distance statistic (0.265–0.282, median 0.273 vs. genuine 0.307–0.336, median 0.326) and exhibited abnormally low launch-to-launch variation (0.183–0.194, median 0.188 vs. genuine 0.323–0.356, median 0.340). Adding an exploratory calibration-derived minimum variation floor to the per-chip p95 rule reduced seven-decoy acceptance to 0/256 (0.0%) while moving genuine acceptance from 228/256 (89.1%) to 225/256 (87.9%; 31/32 on C), though one single-decoy pairing passed 3/32 against GPU C. Code and data are available at https://github.com/JulianBradshaw/gpu-fingerprinting-hardened.

## 1 Introduction: Distinguishing Identification from Authentication

Verifying the physical location of specialized AI accelerators (such as NVIDIA H100, H200, and Blackwell GPUs) is an important building block for compute governance [1]. Delay-based location verification protocols estimate an upper bound on the distance between a verifier and an audited GPU by measuring round-trip signal propagation and computation time against speed-of-light constraints [2]. For such latency bounds to be meaningful, the verifier must authenticate that each challenge response comes from the specific registered GPU rather than a substitute chip or pre-recorded lookup table.

Because hardware-rooted cryptographic attestation keys (such as NVIDIA Confidential Computing [9]) may be vulnerable to invasive physical extraction via focused ion beams [4] or laser side-channel attacks [6], Tee and Happel [10] proposed identifying GPUs via intrinsic manufacturing variations—a software-based Physically Unclonable

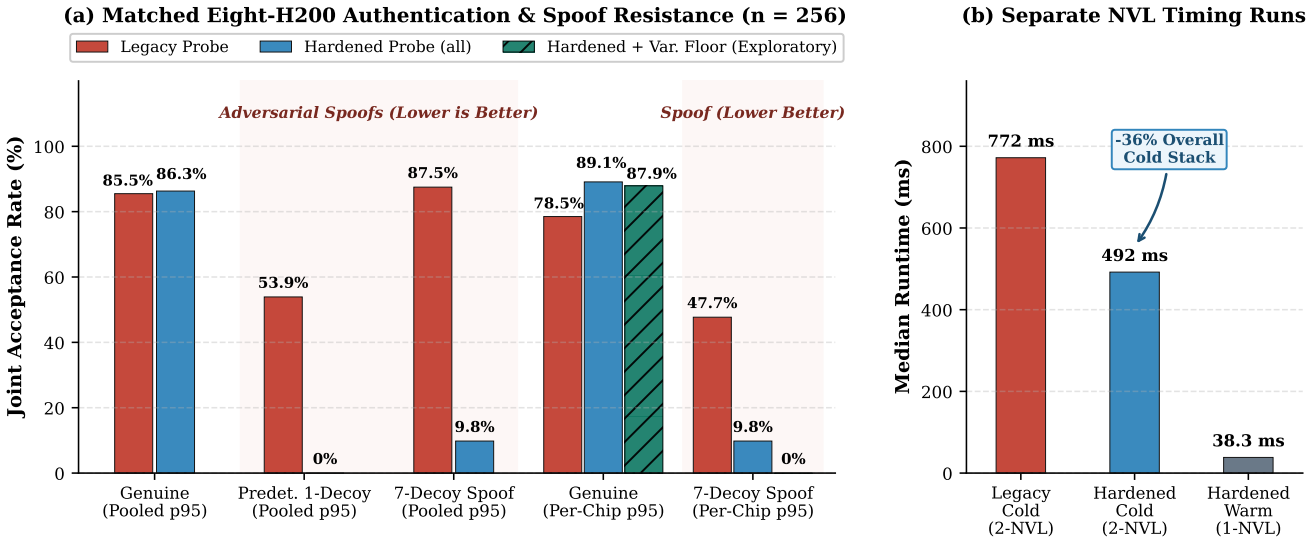



Figure 1: **Matched Eight-H200 Security Comparison and Separate NVL Timing Runs. (a)** On the matched eight-H200 cohort (256 held-out trials per arm), the hardened probe at pooled p95 cuts seven-decoy spoofing from 87.5% (224/256) to 9.8% (25/256) and achieves 0/256 (0.0%) on the predetermined single-decoy assignment (vs. 53.9%, 138/256) while moving genuine acceptance from 85.5% (219/256) to 86.3% (221/256). Under per-chip p95, hardened raises genuine acceptance from 78.5% (201/256) to 89.1% (228/256); an exploratory post-hoc launch-to-launch variation floor (hatched green, evaluated on per-chip p95) reduces seven-decoy acceptance to 0.0% (0/256) at 87.9% (225/256) genuine acceptance. **(b)** In a separate matched two-H200 NVL run, the overall hardened cold implementation ran 36% faster than legacy (492 ms vs. 772 ms); warm 3-launch bursts in a single-H200 NVL pilot took 38.3 ms.

Function (PUF) building on GPU atomic-contention fingerprints [3, 7, 8, 5]. In their proof-of-concept, threads across all Streaming Multiprocessors (SMs) execute a seed-dependent delay loop before racing to increment a shared atomic counter, achieving 98.8% single-run and 100.0% paired-run closed-set re-identification accuracy across 24 H200 GPUs [10].

Tee and Happel explicitly left environmental stability and adversarial security open for future investigation [10]. In particular, closed-set *re-identification* (selecting the nearest registered GPU in a known pool) is necessary for authentication, but does not establish that an adversary cannot synthesize an acceptable response using other GPUs of the same model. Hypothetically, if an adversary could use a small set of decoy GPUs to model and predict the responses of other registered chips, they could attempt to answer challenges for relocated hardware—though an end-to-end location bypass would also have to satisfy live decoy coordination, response synthesis latency, network transport, and verifier deadlines that lie outside offline fingerprint scoring.

In this paper, we evaluate whether atomic-contention GPU fingerprints can support calibrated authentication against replay and decoy-modeling attacks using saved captures, and test a hardened probe implementation on the same physical hardware (Figure 1, Table 1).

## 2 Verification Framework and Attack Models

### 2.1 Registration and Joint Authentication

A verifier registers a cohort of GPUs $\mathcal{G} = \{g_1, \ldots, g_M\}$ by capturing $K = 8$ registration responses $\{R_{g,s}^{(1)}, \ldots, R_{g,s}^{(K)}\}$ per GPU $g$ and challenge $s$. When evaluating a candidate response $Y$ claiming identity $g^*$ on challenge $s$, its score

Table 1: **Overview of Evaluated Fingerprinting Stacks and Matched Eight-H200 Results.** All eight-GPU rates come from the matched cohort (256 held-out trials per arm; 1,792 wrong-chip pairs). The variation floor is an exploratory post-hoc check evaluated on top of the hardened per-chip p95 rule.

| System / Dimension | Legacy Probe [10] | Hardened Probe (`all`) | Exploratory: Hardened + Var. Floor |
|---|---|---|---|
| **Challenge Expansion** | 31-bit LCG (`lcg_state % 64` low-bit period) | 20-round ChaCha20 (256-bit, domain-sep.) | 20-round ChaCha20 (256-bit, domain-sep.) |
| **Contention & Output** | 1 counter + sleeps; 1 raw counter vector | 16 counters, no sleeps; 3 rank bursts | 16 counters, no sleeps; 3 rank bursts |
| **Decision Rule** | Nearest chip + single-launch L1 $\leq \tau$ | 2-of-3 launches nearest + mean L1 $\leq \tau$ | Per-chip p95 + burst variation $\geq v_{\min}$ |
| **Genuine Accept (Pooled p95)** | 85.5% (219/256) | 86.3% (221/256) | *Not evaluated under pooled p95* |
| **Genuine Accept (Per-Chip p95)** | 78.5% (201/256) [80.9% Cal. Max] | 89.1% (228/256) [89.5% Cal. Max] | 87.9% (225/256) [31/32 on GPU C] |
| **Predetermined 1-Decoy Spoof** | 53.9% (138/256) [Pooled p95] | **0.0% (0/256)** [Pooled & Per-Chip p95] | **0.0% (0/256)** [3/32 worst subset E→C] |
| **7-Decoy Ensemble Spoof** | 87.5% (224/256) [Pooled] / 47.7% [Chip] | 9.8% (25/256) [19/25 on C; 19/32 targeting C] | **0.0% (0/256)** [0/32 all tested 2-, 4-, 6-, 7-decoys on C] |
| **Wrong Physical Chip Accepted** | 2.1% (37/1792) [Pooled] / 0.7% [Chip] | 0.9% (17/1792) [Pooled] / 0.8% [Chip] | 0.8% (15/1792) [Per-Chip p95] |
| **Separate NVL Timing Runs** | 772 ms cold (2-H200 NVL matched) | **492 ms cold** (2-NVL) / **38.3 ms warm** (1-NVL) | Same probe execution as Hardened (`all`) |

for chip $g$ is the minimum distance $d(Y,g;s) = \min_k \|Y - R^{(k)}_{g,s}\|_1$. Whereas closed-set identification checks only whether $g^*$ minimizes $d(Y,g;s)$, **joint authentication acceptance** requires:

$$\underbrace{d(Y,g^*;s) < \min_{g \neq g^*} d(Y,g;s)}_{\text{1. Unique Nearest-Chip Match (ties rejected)}} \quad \text{and} \quad \underbrace{d(Y,g^*;s) \leq \tau_{g^*}}_{\text{2. Calibrated Distance Gate}} \tag{1}$$

where the cutoff $\tau_{g^*}$ is estimated strictly from genuine calibration responses captured separately from registration and evaluation. All experiments in this paper score captured responses offline; live challenge state, network deadlines, and hardware attestation lie outside the measurement.

### 2.2 Replay and Decoy Modeling Attacks

1. **Related-Seed Replay:** For the legacy probe, the attacker records responses for base seed $b$ and submits either a single recorded response or the integer-truncated mean of eight recorded responses when challenged on a different seed sharing $b$'s modulo-64 residue.

2. **Single-Decoy and Ensemble Offset Modeling:** For each target/decoy pair $(g^*, d_j)$, the attacker computes the coordinate-wise median difference between their registration means over the 15 other uniform training seeds $\mathcal{S}_{\text{train}}$ (excluding the challenged seed $s^*$):

$$\Delta_{g^*,d_j}[i] = \text{median}_{s \in \mathcal{S}_{\text{train}}} \left( \bar{R}_{g^*,s}[i] - \bar{R}_{d_j,s}[i] \right), \qquad \hat{Y}^{(\text{ensemble})}_{g^*,s^*} = \text{median}_{j \in \{1,\ldots,D\}} \left( Y_{d_j,s^*} + \alpha \cdot \Delta_{g^*,d_j} \right) \tag{2}$$

   where $Y_{d_j,s^*}$ is a held-out evaluation response from decoy $d_j$ on challenge $s^*$, $\alpha \in \{1.0, 1.5, 2.0\}$ is a scaling multiplier, and synthetic outputs are truncated toward zero (legacy) or projected to valid per-round rank permutations (hardened). The predetermined single-decoy assignment selects the next GPU in cyclic cohort order; the seven-decoy ensemble uses all seven other GPUs. *Source-trained* corrections use registration data from the frozen source collection $X$; *destination-trained* corrections use registration data on other seeds from the later collection $Y$.

## 3 Legacy Eight-GPU Assessment

We evaluated the legacy kernel across four complete collections (`Run 1`–`Run 4`) on 7 November 2026 using the same eight plain H200 UUIDs in identical order over approximately 13 minutes across two rental instance IDs (`59660701` for Runs 1–2 at 8:47:06 p.m. and 8:51:46 p.m. MT; `59661478` for Runs 3–4 at 8:56:41 p.m. and 9:00:27 p.m. MT). Each collection contains 32 seeds and 10 measurements per GPU/seed (2,560 fingerprints per collection, 10,240 total): 16 uniform 31-bit seeds and four modulo-64 residue groups ($b, b+64, b+2^{16}, b+2^{24}$ for $b \in \{0,1,2,3\}$). The default split assigns runs 0–7 to registration, run 8 to evaluation, and run 9 to calibration. Figure 2 and Table 2 summarize the legacy findings.

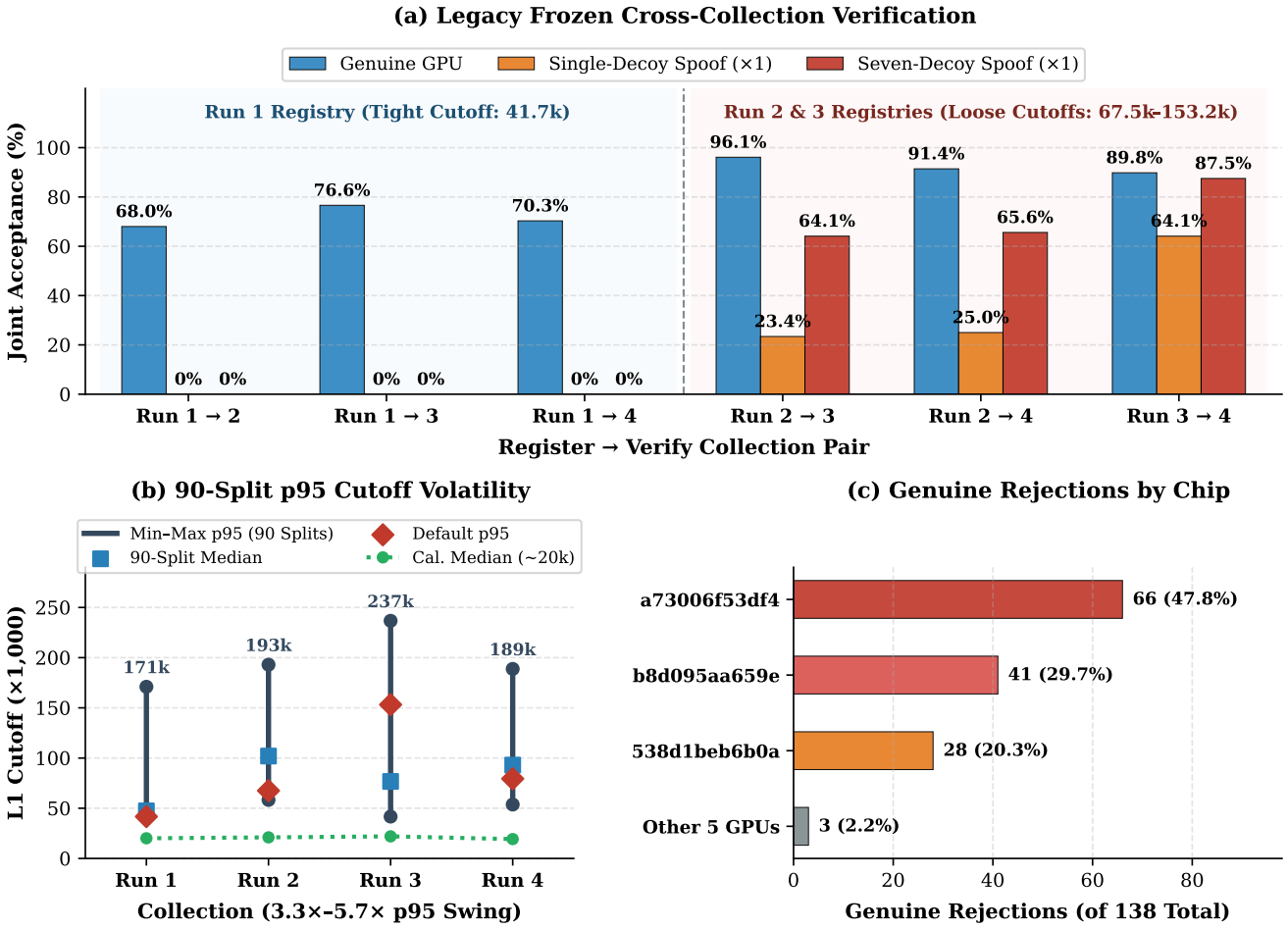



Figure 2: **Legacy Probe Behavior Across Collections and Sample Splits. (a)** While nearest-chip identification without a cutoff stayed at 94.5%–100.0%, frozen joint acceptance ($X \to Y$) varied sharply: Run 1's cutoff (41.7k) rejected 23.4%–32.0% of later genuine responses, whereas Run 2 and Run 3 cutoffs admitted 64.1%–87.5% of 7-decoy spoofs. **(b)** Across all 90 calibration/evaluation split choices within each collection, calibration medians remained $\sim$ 19.3k–22.0k while p95 cutoffs varied by 3.3× to 5.7× (41.7k–236.7k). **(c)** Three of eight H200 chips accounted for 97.8% (135/138) of genuine forward rejections.

1. **Low-Bit Seed Structure and Related-Seed Replay:** The legacy kernel initializes an LCG with `seed` $\oplus$ `smid`, masks updates to 31 bits, and selects a racing thread via `lcg_state` mod 64. Because the low six bits evolve independently of higher bits, seeds congruent modulo 64 produce identical thread-selection sequences on each SM, while higher bits influence the delay loop (100,000 + `lcg_state` mod 17). Single-H200 measurements gave mean L1 distances of 12,385 for repeated runs of the same seed, 116,620 for distinct seeds congruent modulo 64, and 1,875,854 for seeds with different residues. Under Run 3's wider p95 cutoff (153,213.9), recorded related-seed replay passed 91.7% of attempts (83.3% for mean-of-eight replay; Table 2).

2. **Identification vs. Frozen Calibration Tradeoffs:** Across all 12 directed cross-collection comparisons, nearest-chip accuracy without a distance threshold remained 94.5%–100.0%. However, frozen joint acceptance (Figure 2a, Table 2) dropped to 68.0%–76.6% from Run 1, while Run 2 and Run 3 registries admitted 64.1%–87.5% of source-trained ×1 seven-decoy spoofs. In a retrospective sweep across all score breakpoints, the smallest single cutoff accepting $\geq$ 95% of genuine responses in every later collection admitted 100.0%, 75.0%, and 87.5% of worst source-trained spoofs (93.0%, 86.7%, and 99.2% destination-trained) for Runs 1–3. Conversely, the largest cutoff rejecting all tested spoofs left minimum genuine acceptance at 68.0%,

Table 2: **Legacy Probe Within-Collection and Frozen Cross-Collection Verification.** For $3 \rightarrow 4$, predetermined cyclic subsets passed at 62.5%–64.1% (1 decoy), 51.6%–67.2% (2 decoys), 82.8%–87.5% (4 decoys), and 87.5% (7 decoys).

| **Within-Collection Pooled p95 Performance ($n = 128$ / 96 replay)** | | | | | | **Frozen Cross-Collection Verification ($X \rightarrow Y$) & Bounds** | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Run** | **p95 Cutoff** | **Genuine** | **1-Decoy** | **7-Decoy** | **Replay (Rec/Mean8)** | **Pair** | **Genuine** | **1-Decoy** | **7-Decoy** | **Spoof @ $\geq$ 95% Gen (Src/Dest)** |
| Run 1 | 41,725.5 | 92.2% | 0.0% | 8.6% | 0.0% / 0.0% | $1 \rightarrow 2/3/4$ | 68.0 / 76.6 / 70.3% | 0.0% | 0.0% | 100.0% / 93.0% (@ 473k) |
| Run 2 | 67,486.9 | 92.2% | 24.2% | 68.0% | 1.0% / 0.0% | $2 \rightarrow 3$ | 96.1% | 23.4% | 64.1% | 75.0% / 86.7% (@ 114k) |
| Run 3 | 153,213.9 | 95.3% | 68.0% | 87.5% | 91.7% / 83.3% | $2 \rightarrow 4$ | 91.4% | 25.0% | 65.6% | [Run 2 bound covers 3, 4] |
| Run 4 | 79,429.9 | 96.9% | 28.9% | 75.0% | 3.1% / 3.1% | $3 \rightarrow 4$ | 89.8% | 64.1% | 87.5% | 87.5% / 99.2% (@ 293k) |

83.6%, and 80.5%.

3. **Split Sensitivity, Chip Concentration, and Simple Defenses:** Across all 90 ordered choices of calibration and evaluation runs within a collection, p95 spanned 41.7k–171.0k (Run 1, median 47.4k), 58.3k–193.0k (Run 2, median 102.0k), 41.7k–236.7k (Run 3, median 76.7k), and 53.7k–188.8k (Run 4, median 93.0k), while calibration medians stayed near 19.3k–22.0k (Figure 2b). Across 10 balanced cyclic splits (30 source/split configurations), $20/30$ could reach 95% genuine acceptance in later collections, with lowest achievable worst $\times 1$ spoof rates spanning 14.1%–99.2% (the other $10/30$ failed due to identification errors even without a distance gate). Three UUID suffixes (`a73006f53df4`: 66, `b8d095aa659e`: 41, `538d1beb6b0a`: 28) accounted for $135/138$ (97.8%) of genuine forward rejections (Figure 2c). Source-trained chip normalization improved $2 \rightarrow 4$ (92.2% genuine, 12.5% spoof vs. $91.4\%/68.8\%$ pooled) but left $3 \rightarrow 4$ at 89.1% genuine and 73.4% spoof (and chip-specific cutoffs on $3 \rightarrow 4$ gave 84.4% genuine and 23.4% spoof).

4. **Legacy Cold Timing and Sleep Removal:** Thirty single-H200 cold process runs averaged 848.6 ms (SD = 98.8 ms, range 764.7–1,345.0 ms, p95 − p5 = 120.6 ms). Removing reset-round sleeps reduced median process time by $\sim$ 42.9 ms and shifted one tested fingerprint by L1 180,086 (versus 11,624 genuine repeat distance).

# 4 Hardened Probe Evaluation and Exploratory Follow-Up

## 4.1 Core Hardened Architecture

The hardened probe modifies four core components of the fingerprinting stack:

1. **Domain-Separated 20-Round ChaCha20 Expansion:** Expands a public 256-bit challenge using the 20-round ChaCha20 block function with separate SM and round domains, providing diffusion without a secret key.

2. **Sleep-Free Delay Work & 16-Address Contention:** Replaces deliberate sleep calls with challenge-dependent delay work and synchronization across 16 counter addresses while recording the physical SM map.

3. **Per-Round SM Rank Permutations:** Emits per-round rank permutations across active SMs in either the `all` profile (all 132 mapped SMs) or the `subset` profile (4–8 active SMs per round).

4. **Two-of-Three Warm Micro-Burst Voting:** Captures three launches per burst inside a single warm process (8 registration bursts = 24 templates). Each launch is scored by its minimum mean active-coordinate L1 distance to the claimed GPU's 24 templates; joint acceptance requires at least **2 of 3 launches** to be uniquely nearest to the claimed GPU and within its calibration cutoff.

### 4.2 Matched Eight-H200 Ensemble Benchmark (and NVL Timing Pilots)

We first verified engineering consistency and timing on H200 NVL hosts (8–9 November 2026). On a single H200 NVL (rental ID `59832716`, 388 bursts across a 132-SM map), both `all` (61/64) and `subset` (58/64) met the prespecified 58/64 cross-process consistency screen, with median warm probe times per 3-launch burst of 38.29 ms (`all`) and 36.18 ms (`subset`), excluding process startup and transport. On a two-H200 NVL pilot (`59835170`, 776 bursts, evaluated 10–11.5 min after calibration freeze), both modes achieved 128/128 nearest-chip accuracy, 0/128 wrong-chip accepts, and 123/128 (`all`) to 127/128 (`subset`) per-chip p95 genuine acceptance (127/128 and 128/128 at calibration max). In a matched alternating-order two-H200 NVL run (`59934299`, 516 responses/arm, 11.5–13 min freeze), hardened achieved 123/128 (96.1%) genuine per-chip p95 acceptance versus 118/128 (92.2%) for legacy (126/128 vs. 123/128 at calibration max). Both implementations accepted 0/64 bit-6 cross-challenge replay attempts. Hardened ranked $\times 1$ single-decoy predictions first on 0/128 trials (vs. 128/128 for legacy prior to the cutoff), and reduced median cold local process runtime (start to exit, including CUDA setup and stdout) from 772 ms to 492 ms ($-36\%$, faster in all 8/8 paired attempts; Figure 1b). This 36% reduction reflects the measured implementation as a whole rather than isolating sleep removal alone.

Our main security comparison (9 November 2026, rental ID `59951659`) evaluated legacy and hardened `all` head-to-head on a separate cohort of eight plain H200 GPUs on one host across 16 challenge slots (8 registration, 3 calibration, and 2 evaluation responses per GPU/challenge $= 1{,}728$ captures/arm including smoke; held-out capture began 300–396 s after sealing calibration). Hardened decoy predictions were projected to valid per-round rank permutations and scored under the standard two-of-three rule. As shown in Figure 1a and Table 1:

- **Pooled p95 Comparison:** Seven-decoy acceptance fell from 224/256 (87.5%) in Legacy to 25/256 (9.8%) in Hardened (a 77.7 percentage-point reduction), while acceptance of the predetermined single-decoy assignment fell from 138/256 (53.9%) to 0/256 (0.0%). Genuine acceptance was 219/256 (85.5%) for Legacy versus 221/256 (86.3%) for Hardened (nearest-chip accuracy 219/256 [85.5%] vs. 232/256 [90.6%]; wrong-chip acceptance 37/1,792 [2.1%] vs. 17/1,792 [0.9%]).
- **Per-Chip p95 and Calibration Max:** Under per-chip p95 (the 47th of 48 calibration scores per chip), genuine acceptance was 201/256 (78.5%) for Legacy versus 228/256 (89.1%) for Hardened, while seven-decoy acceptance was 122/256 (47.7%) versus 25/256 (9.8%) and wrong-chip acceptance was 12/1,792 versus 15/1,792. At each chip's calibration maximum, genuine acceptance was 207/256 (80.9%) versus 229/256 (89.5%), and seven-decoy acceptance was 181/256 (70.7%) versus 25/256 (9.8%). Legacy also selected the wrong nearest GPU on all 32/32 held-out responses from GPU E before applying any cutoff, whereas Hardened accepted 20/32 genuine bursts on GPU E under per-chip p95.

### 4.3 Exploratory Offline Follow-Up: Decoy Subsets and Launch Variation on GPU C

At the hardened per-chip p95 cutoff, GPU C accounted for 19 of the 25 accepted seven-decoy responses cohort-wide and accepted 19/32 attempts targeting C (while accepting 32/32 genuine bursts); GPUs E and G accounted for the remaining 2 and 4 accepted spoofs (2/32 and 4/32 targeting E and G). To investigate GPU C's vulnerability, we performed an exploratory offline sweep across all 71 decoy subsets of sizes $D \in \{1,2,4,6,7\}$ targeting GPU C under its frozen per-chip p95 cutoff (Figure 3b, Table 3). While the predetermined single-decoy assignment had passed 0/32, the strongest individual decoy in the exhaustive sweep (GPU E) passed 3/32 on GPU C, the strongest two-decoy pair (GPUs B and F) and four-decoy subsets passed up to 26/32, and removing any single card from the seven-decoy ensemble still left 18/32–24/32 accepts.

Examining the score structure on GPU C (Figure 3a) revealed two empirical observations:

- **Burst distance scores below genuine responses:** The verifier accepts a burst when at least two launches each identify the claimed GPU as uniquely nearest and clear its cutoff.

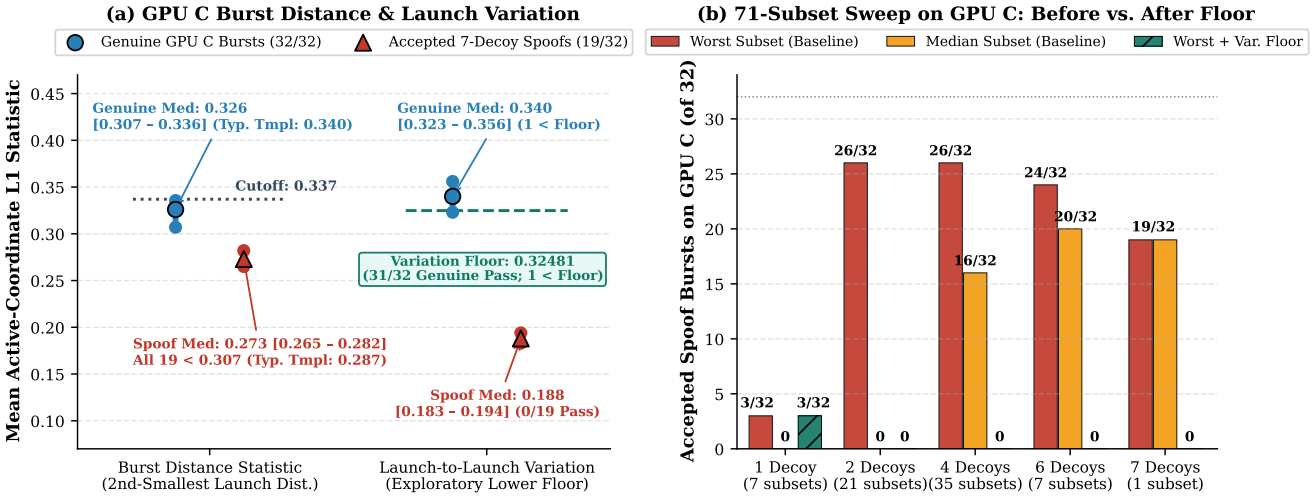



Figure 3: **Exploratory Analysis of GPU C Score Structure and Candidate Variation Floor. (a)** On GPU C, all 19 accepted seven-decoy bursts scored below every one of GPU C's 32 genuine bursts on the burst distance statistic (the second-smallest of the three launch distances: spoof range 0.265–0.282, median 0.273 vs. genuine range 0.307–0.336, median 0.326; GPU C's per-chip p95 upper cutoff was 0.337, and median distances to typical registration templates were 0.287 vs. 0.340). However, accepted spoofs exhibited much lower launch-to-launch variation across the three launches in a burst (0.183–0.194, median 0.188) than genuine bursts (0.323–0.356, median 0.340, with 1 genuine burst at 0.323 falling below GPU C's calibration variation floor of 0.32481). **(b)** In offline rescoring across all 71 tested decoy subsets on GPU C ($D \in \{1,2,4,6,7\}$), adding the minimum variation floor reduced all tested 2-, 4-, 6-, and 7-decoy subsets to $0/32$, while single decoy GPU E still passed $3/32$.

Table 3: **Exploratory 71-Subset Decoy Sweep on Target GPU C ($D \in \{1,2,4,6,7\}$) and Offline Evaluation of Candidate Lower Bounds (Hardened Per-Chip p95 Baseline).**

| Exhaustive 71-Subset Sweep on Target GPU C (of 32) | | | | Exploratory Lower-Bound Rules Across All 8 GPUs | | | |
|---|---|---|---|---|---|---|---|
| **Decoys ($D$)** | **Subsets** | **Min / Med / Max** | **Max w/ Var. Floor** | **Hardened Rule (Per-Chip p95)** | **Genuine (8 GPUs)** | **7-Decoy (8 GPUs)** | **Worst 2-Decoy (C)** |
| 1 | 7 | 0 / 0 / 3 | 3/32 (GPU E) | Per-chip p95 baseline | 228/256 (89.1%) | 25/256 (9.8%) | 26/32 (81.3%) |
| 2 | 21 | 0 / 0 / 26 | **0/32** | + Min own-distance floor | 221/256 (86.3%) | **0/256 (0.0%)** | 21/32 (65.6%) |
| 4 | 35 | 8 / 16 / 26 | **0/32** | **+ Min variation floor** | **225/256 (87.9%)** | **0/256 (0.0%)** | **0/32 (0.0%)** |
| 6 | 7 | 18 / 20 / 24 | **0/32** | *Note: Lower floors use genuine calibration only (C's variation floor = 0.32481).* | | | |
| 7 | 1 | 19 / 19 / 19 | **0/32** | *Retains 31/32 genuine bursts on C and keeps wrong-chip accepts at 15/1,792.* | | | |

The joint score is the second-smallest distance among uniquely matching launches. Figure 3a plots the second-smallest distance across all three launches. On this measure, all 19 accepted seven-decoy bursts scored lower than every one of GPU C's 32 genuine bursts: accepted spoofs spanned 0.265–0.282 (median 0.273), whereas genuine GPU C bursts spanned 0.307–0.336 (median 0.326, evaluated against GPU C's per-chip p95 cutoff of 0.337). The ensemble was also closer to typical registration templates (median distance 0.287 vs. 0.340). The lowest joint score among accepted spoofs was 0.265.
An upper cutoff that rejects all seven-decoy spoofs would also reject all 32 genuine GPU C bursts.

- **Low launch-to-launch variation within spoofed bursts:** Measuring a burst's internal variation as the median of its three pairwise mean L1 distances, accepted seven-decoy spoofs on GPU C ranged from 0.183 to 0.194 (median 0.188), compared with 0.323 to 0.356 (median 0.340) for genuine GPU C bursts. GPU C's calibration minimum variation floor is 0.32481, which rejects all 19 seven-decoy spoofs while rejecting 1 of GPU C's 32 genuine bursts (the minimum genuine burst at 0.323, leaving $31/32$ genuine C bursts accepted). A plausible explanation is that taking the coordinate-wise median across multiple decoys dampens single-chip launch-to-launch jitter, though the experiment did not isolate the underlying mechanism.

Motivated by these observations from the same pilot, we evaluated two exploratory lower-bound rules fitted strictly on each target's smallest genuine calibration value on top of the per-chip p95 baseline (Table 3, right): rejecting bursts whose own-distance statistic fell below the calibration minimum, and rejecting bursts whose launch-to-launch variation fell below the calibration minimum. While the minimum-distance floor blocked the full seven-decoy ensemble ($0/256$), it still admitted $21/32$ two-decoy spoofs on GPU C. The minimum-variation floor reduced seven-decoy acceptance to $0/256$ across all eight GPUs and blocked every tested 2-, 4-, 6-, and 7-decoy subset on GPU C ($0/32$; sizes 3 and 5 were omitted from the sweep), while moving genuine acceptance from $228/256$ (89.1%) to $225/256$ (87.9%; $31/32$ on GPU C) and leaving wrong-chip acceptance unchanged at $15/1{,}792$. Because this rule was formulated after inspecting the pilot data, it requires prospective validation.

## 5 Limitations, Replication Requirements, and Follow-Ups

The reported rates are descriptive; trials reuse GPUs, challenges, and registration templates across evaluations, and the historical split sweeps reuse stored captures rather than representing independent statistical trials. Three primary follow-up directions remain:

1. **Prospective Validation of the Variation Floor Against Adaptive Spoofs:** Freeze the launch-to-launch variation statistic and calibration floor prior to fresh data collection, and test against adaptive spoofs designed to imitate genuine launch-to-launch variation while remaining below the upper distance cutoff.
2. **Longer-Gap Stability and Separate-Host Replication:** All captures in this study used short gaps (5–13 minutes) on single hosts without controlled power cycles, transport, or separate target and decoy hosts. Future experiments should evaluate stability across restarts and longer sessions, separate target and decoy hosts, and collect independent attacker profiles.
3. **Component Ablations and End-to-End Location Verification:** Because the legacy and hardened probes use different native response budgets, matched-budget ablations are needed to isolate the individual contributions of ChaCha20 expansion, 16-address contention, rank formatting, and burst voting. Practical location verification requires separate experiments incorporating verifier-side network timing deadlines, live challenge state, and trusted hardware attestation [9].

**Replication Checklist:**

- **Hardware & Environment Manifest:** Exact H200 SKU per archive (plain H200 vs. H200 NVL, 132-SM map), CUDA version (12.4), host OS kernel, and topology.
- **CUDA Kernel Source, Build Flags & Native Response Sizes:** Full source for the legacy harness and hardened probe (`all` and `subset`), exact `nvcc` compilation flags used in the archived eight-GPU runner (`-arch=sm_90`, with hardened compiled at `-O2` and legacy compiled at `-O0`), native response sizes (1 launch vs. 3-launch burst of rank permutations), 20-round ChaCha20 domain constants, 16 counter address offsets, and challenge-dependent delay parameters.
- **Scoring & Projection Scripts:** Python scripts (`frozen_registry_tradeoff.py`, `extended_offline_analysis.py`, `eight_h200_ensemble.py`) implementing rank-permutation projection, interpolated vs. order-statistic p95 cutoffs, and pairwise burst variation.
- **Archived Datasets & SHA-256 Checksums:** Raw archives and JSON summaries for legacy collections (`59660701`, `59661478`) and hardened/matched runs (`59832716`, `59835170`, `59934299`, `59951659`) with SHA-256 manifests.

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